Avi Chauhan

Baton Rouge, LA | chauhanavi2311@gmail.com | 332-203-6732 | chauhanavi.com | linkedin.com/in/developeravichauhan | github.com/chauhanavi21

Technical Skills

Languages Python, JavaScript, TypeScript, SQL, C++, C#, Go, Bash
CS Fundamentals Data Structures, Algorithms, Object Oriented Design, Debugging, Code Review
AI Assisted Development GenAI Coding Assistants, LangChain, LLM Tool Calling, Prompt Engineering
Databases PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, pgvector
Cloud & DevOps AWS EC2, S3, Lambda, EKS, Microsoft Azure, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD
Backend FastAPI, Node.js, REST APIs, Celery, WebSockets
Version Control & Tools Git, GitHub Actions, Jenkins, Jira, Postman, OpenTelemetry
Observability Prometheus, Grafana

Work Experience

IBM Consulting — On-site

Associate Application Developer — *Sep 2026 to Present*

- Partnered with an enterprise client on a confidential two week engagement to migrate the client's software from its existing legacy on premises structure onto the Microsoft Azure platform, scoping the migration requirements against the client's current architecture.
- Re-architected core application components for Azure compatibility, replacing legacy infrastructure dependencies with Azure native services to support a more scalable, cloud first deployment for the client.
- Identified and resolved legacy workflow bottlenecks uncovered during the migration, streamlining the client's operational processes post migration to increase overall system efficiency.

Silent Infotech Pvt Ltd — On-site

Software Engineer — *Apr 2022 to May 2024*

- Engineered a production RAG pipeline over 5,000+ internal documents using FastAPI, Sentence Transformers and hybrid FAISS and BM25 retrieval, serving semantic Q&A to a 20+ engineer team and reducing manual document lookup time by over 60%.
- Optimized model serving via ONNX export and INT8 quantization, reducing inference latency by 38% while maintaining accuracy thresholds for production deployment on resource constrained servers.
- Designed an end to end data pipeline covering ingestion, chunking, embedding, retrieval and evaluation at scale for the production RAG system.
- Measured answer relevance using a faithfulness scoring methodology and outperformed the baseline LLM by 38% on internal benchmarks.
- Containerized the full pipeline with Docker and deployed via AWS Lambda and S3, then configured GitHub Actions CI/CD across 4 pipeline stages, reducing release overhead and enabling zero touch deployments on every merge.

Opus Technologies — Remote

Software Engineering Intern — *Sep 2021 to Feb 2022*

- Fine-tuned VGG-16 on a 100K+ image dataset across 5 disease classes for medical image classification, lifting test accuracy from 71% to 92% by applying targeted augmentation and class-weighting on severely imbalanced data.
- Developed preprocessing and evaluation pipelines with Pandas, NumPy and scikit-learn, streamlining hyperparameter tuning across 15+ model configurations and reducing experiment cycle time by 3x.
- Delivered a modular Python codebase with documented APIs across 3 classification modules, enabling the team to onboard 2 additional disease datasets with under 4 hours of integration work each.

Projects

CLAWD — 2025

Python, LangChain, LLM Tool Calling, File System Agents, CLI

- Built a local AI coding agent that plans, edits and tests code changes across a repository through LLM driven tool calling, applying sound judgement about when to trust the model's proposed changes before committing them.
- Implemented a file system interaction layer letting the agent read, write and execute commands autonomously across the target repository.

– Added session memory to support multi step reasoning across sequential coding tasks, then open sourced the project on GitHub for community use and iteration.

Developer Platform Portal GitOps 2025

Backstage, TypeScript, Terraform, AWS EKS, ArgoCD, GitHub Actions, Grafana

– Launched an internal developer portal on Backstage cataloging 8+ microservices with golden path templates for standardized service scaffolding.
– Configured per service GitHub Actions CI pipelines deployed to AWS EKS via ArgoCD GitOps for continuous, automated delivery.
– Provisioned the underlying infrastructure with Terraform and built OpenTelemetry fed Grafana dashboards for platform wide observability.

Enterprise AI Knowledge Copilot 2025

FastAPI, Next.js, PostgreSQL, Redis, Celery, pgvector, Langfuse, Ragas, Docker

– Architected a self hosted internal AI platform supporting 3+ document connectors and 1,000+ indexed knowledge chunks over a Next.js chat UI and FastAPI backend.
– Built an async ingestion pipeline with Celery and Redis feeding pgvector backed semantic retrieval for the platform's RAG layer.
– Integrated Langfuse to trace 100% of LLM calls with prompt versioning and cost tracking, then established Ragas evaluation pipelines achieving a faithfulness score of 0.86 on internal test sets.

Local-First Collaborative Whiteboard CRDT 2025

React, TypeScript, Yjs, Canvas API, Node.js, WebSockets, PostgreSQL

– Wired a real time multi user whiteboard supporting 10+ concurrent collaborators using Yjs CRDT shared types over a y-websocket server, surfacing live presence as a distinct awareness layer with sub-100ms sync latency.
– Established offline first persistence with IndexedDB, retaining 100% of local edits through disconnections and merging conflict free on reconnect.
– Backed the merge layer with PostgreSQL snapshots enabling full session restore across unlimited shareable room links.

Education

State University of New York at Albany Albany, NY
Master of Science, Computer and Information Sciences, GPA 3.4 *Aug 2024 to May 2026*

Indus University Ahmedabad, India
Bachelor of Engineering, Computer Science and Engineering, GPA 3.7 *Sep 2020 to May 2024*

Certifications

AWS Certified Solutions Architect Associate | Validation 222d1eb066f24407bb04abedb0c10fee
AWS Certified AI Practitioner | Validation 5289992c1c274ebeafa5aaddeb0c5d35
IBM Generative and Agentic AI Foundation, IBM Consulting Foundational Course